Piracy and PIPA/SOPA

This is our final project for Decision Making Under Uncertainty

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Abstract

We investigated whether political funding is associated with U.S. legislators stances on PIPA/SOPA using the piracy dataset.

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Introduction

PIPA and SOPA were proposed laws intended to address online piracy, but they also created major debate about internet regulation, copyright enforcement, and political influence. In this project, we investigate whether political funding is associated with legislators stances on PIPA/SOPA. To answer this question, we use the `piracy` dataset from openintro.org, which includes information about U.S. legislators, their party, state, chamber, years in office, funding received from pro- and anti-PIPA/SOPA groups, and their stance on the issue.

Project Motivation

This question matters because political decisions may be influenced by many factors, including party affiliation, ideology, public opinion, and campaign funding. Finding the factors at play helps us understand how political decisions can be influenced.

To study this, we compare pro-PIPA/SOPA funding and anti-PIPA/SOPA funding for each legislator. In our analysis, we calculate a funding difference variable using `money_pro - money_con`, where positive values mean that a legislator received more pro-PIPA/SOPA funding than anti-PIPA/SOPA funding. We then use exploratory plots and a linear regression model to test whether this funding difference, or whether any other factors are associated with stance.

Discussion

Data Source and Variables

Source: The dataset used in this project is called "piracy" and is available as an R dataset. It contains information about U.S. politicians including their party, funding, and stance on an issue.

Context: This dataset represents political data for members of the House and Senate. It includes information about how much money politicians receive (supporting and opposing), their political party, and their stance (yes, no, or unknown).

Population vs Sample: The dataset represents the population of legislators included in this study because it contains all legislators recorded in the dataset. However, for the statistical analyses, the subset of legislators with clear "yes" or "no" stances after removing "unknown" responses can be treated as the sample used for analysis.

Size of Dataset:

- Number of observations: 534
- Number of variables: 8

Variables Description:

- name: Name of politician
- party: Political party (D, R, I)
- state: State represented
- money_pro: Money supporting
- money_con: Money opposing
- years: Years in office
- stance: Position (yes, no, etc.)
- chamber: House or Senate

Data Cleaning and Preparation

```r
# Load data
data(piracy)


data <- piracy
# View data
head(data)
```

name	party	state	money_pro	money_con	years	stance	chamber
Ackerman, Gary	D	NY	13350	14800	30	unknown	house
Adams, Sandra	R	FL	3500	5650	2	unknown	house
Aderholt, Robert	R	AL	4779	23944	16	unknown	house
Akin, Todd	R	MO	2500	8200	12	no	house
Alexander, Rodney	R	LA	3500	2700	10	no	house
Altmire, Jason	D	PA	24250	10650	6	unknown	house

```r
# Size
dim(data)
```

```
## [1] 534   8
```

```r
# Structure
str(data)
```

```
## tibble [534 x 8] (S3: tbl_df/tbl/data.frame)
##  $ name     : chr [1:534] "Ackerman, Gary" "Adams, Sandra" "Aderholt, Robert" "Akin, Todd"
##  $ party    : Factor w/ 3 levels " D"," R"," I": 1 2 2 2 2 1 2 2 1 2 ...
##  $ state    : Factor w/ 50 levels "AK","AL","AR",..: 34 9 2 24 18 38 22 33 31 35 ...
##  $ money_pro: num [1:534] 13350 3500 4779 2500 3500 ...
##  $ money_con: num [1:534] 14800 5650 23944 8200 2700 ...
##  $ years    : int [1:534] 30 2 16 12 10 6 2 2 24 4 ...
##  $ stance   : Factor w/ 5 levels "leaning no","no",..: 4 4 4 2 2 4 2 5 2 4 ...
##  $ chamber  : Factor w/ 2 levels "house","senate": 1 1 1 1 1 1 1 1 1 1 ...
```

```r
# Summary
summary(data)
```

```
##      name              party        state        money_pro         money_con
##  Length:534          D:243   CA     : 55   Min.   : -5000   Min.   : -1000
##  Class :character    R:289   TX     : 34   1st Qu.:  4500   1st Qu.:  4500
##  Mode  :character    I:  2   NY     : 31   Median : 11700   Median : 10200
##                              FL     : 27   Mean   : 26326   Mean   : 23193
##                              IL     : 21   3rd Qu.: 27462   3rd Qu.: 23947
##                              PA     : 21   Max.   :571600   Max.   :550000
##                              (Other):345   NA's   :14       NA's   :35
##      years              stance      chamber
##  Min.   : 1.00   leaning no: 44   house :434
##  1st Qu.: 4.00   no        :122   senate:100
##  Median :10.00   undecided : 11
##  Mean   :11.76   unknown   :294
##  3rd Qu.:18.00   yes       : 63
```

```
##  Max.   :58.00
##
```

```r
# Missing values
colSums(is.na(data))
```

```
##       name      party      state money_pro money_con      years     stance    chamber
##          0          0          0        14        35          0          0          0
```

```r
# Fix spaces
data$party <- trimws(data$party)

# Convert to factors
data$party <- as.factor(data$party)
data$state <- as.factor(data$state)
data$stance <- as.factor(data$stance)
data$chamber <- as.factor(data$chamber)

# Clean data
data_clean <- na.omit(data)

# Counts
table(data$party)
```

```
##
##   D   I   R
## 243   2 289
```

```r
table(data$stance)
```

```
##
## leaning no          no  undecided    unknown        yes
##         44         122         11        294         63
```

```r
table(data$chamber)
```

```
##
```

```
##  house senate
##    434    100
```

Research Questions and Hypothesis

Research Question

Our primary research question is: **Is funding associated with legislators stances on PIPA/SOPA?**

We came up with this question because campaign funding is often discussed as a possible influence on political decisions. Since the dataset includes both funding information and legislator stance, we can test whether there is a relationship between pro- and anti-PIPA/SOPA funding and whether a legislator supported the bill.

Our Hypothesis

We hope to model this relationship:

$$stance = \beta_0 + \beta_1(difference) + \beta_2(partyaffiliation) + \epsilon$$

Our hypotheses are:

$$H_0(1) : \beta_1 = 0$$

$$H_A(1) : \beta_1 \neq 0$$

The null hypothesis H_0 is testing when difference of funding pro vs con has no slope. The alternative hypothesis H_A is testing when the difference slope is not 0.

$$H_0(2) : \beta_2 = 0$$

$$H_A(2) : \beta_2 \neq 0$$

The null hypothesis H_0 is testing when party affiliation is not associated with stance. The alternative hypothesis H_A is testing that party affiliation is associated with stance.

Type I Error: Concluding money associates with stance when it doesn't **Type II Error**: Concluding there is no association when there is association **Worse Type I**: — falsely accusing legislators of being influenced by money

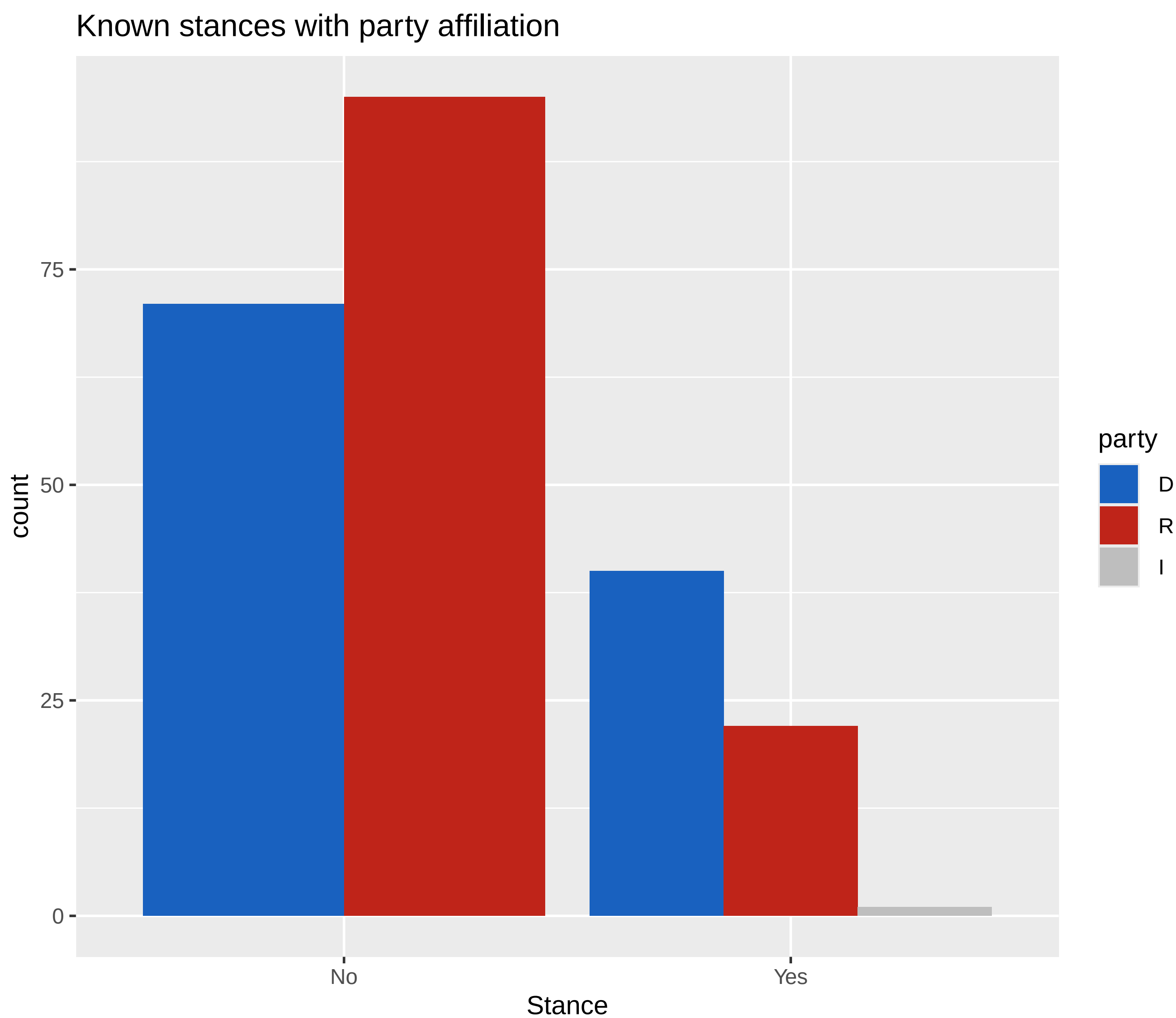
Main Analyses

Exploratory Data Analysis

Bar chart for party relation to stance

The plot shows how many legislators are opposed to and supportive of PIPA/SOPA, separated by their party affiliations. It helps to get an understanding of the party affiliations and how they might potentially correlate with stances.

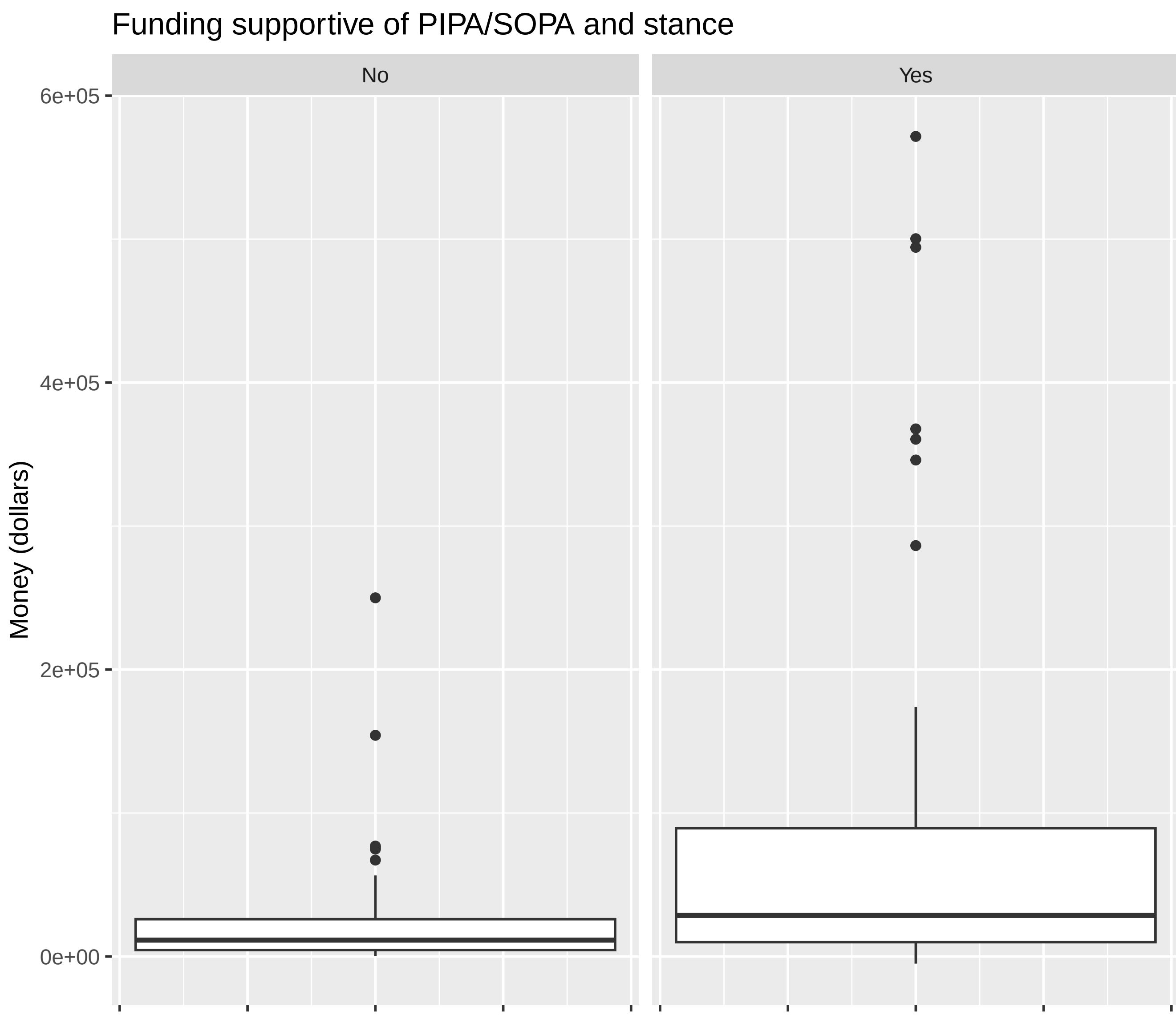
There appears to be some correlation between the two, with more Republicans being opposed to the bills and more Democrats being in support of them. This fact might add some additional variance to our results, and so could explain it.

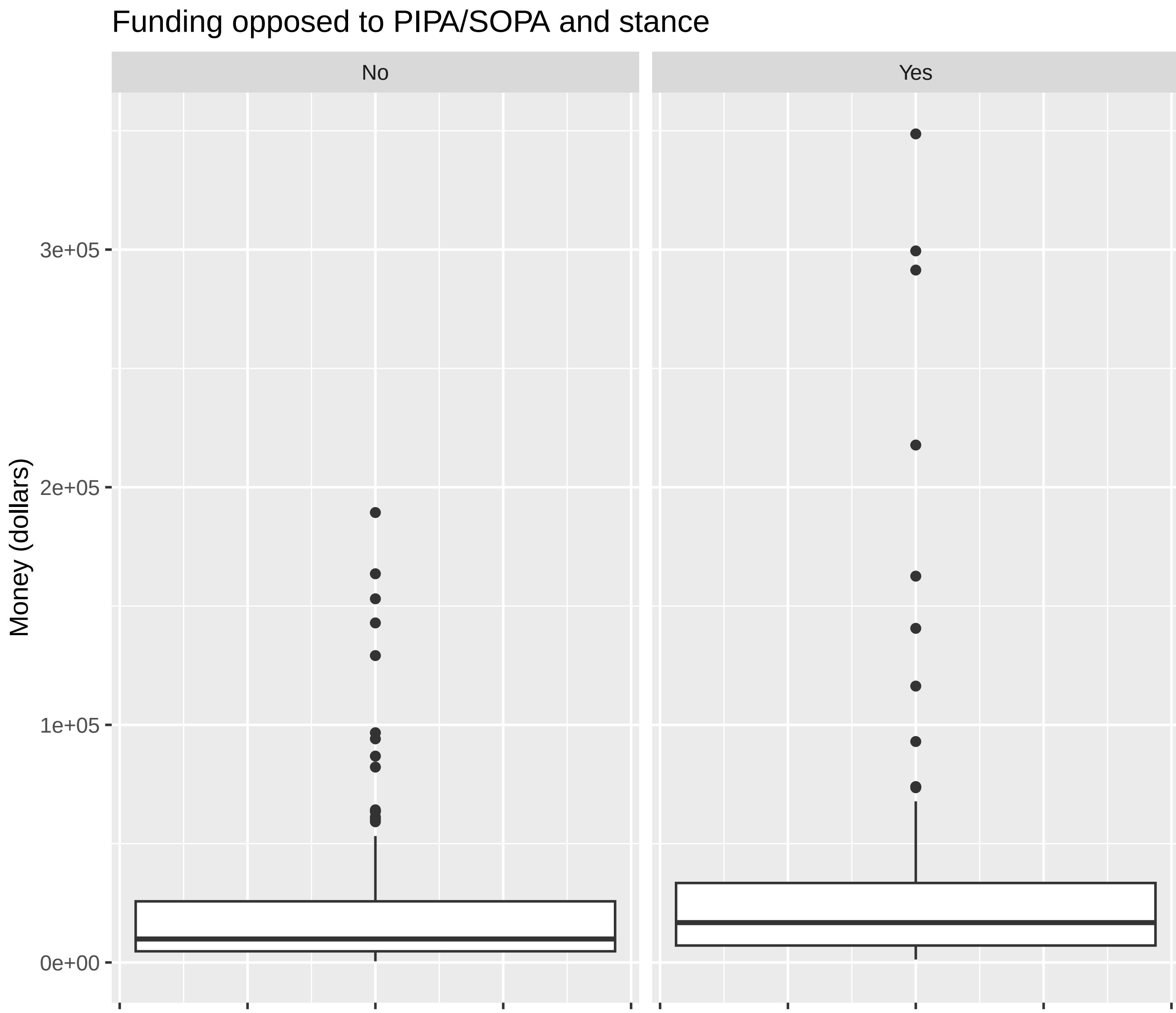


Box plots for funding relation to stance

The two plots show how much funding legislators received from both pro- and anti-PIPA/SOPA sources, seperated by whether or not the legislators themselves support or oppose the bills.

These plots help to visually illustrate the difference in both types of funding between those opposed and supportive of the bills. It can be clearly seen that those who support the bills also received a significant amount more funding from pro-PIPA/SOPA donors. This suggests a strong correlation between the two. Notably, there is no parallel to this for funding from anti-PIPA/SOPA donors—those who oppose it didn't receive much more funding than those who support it

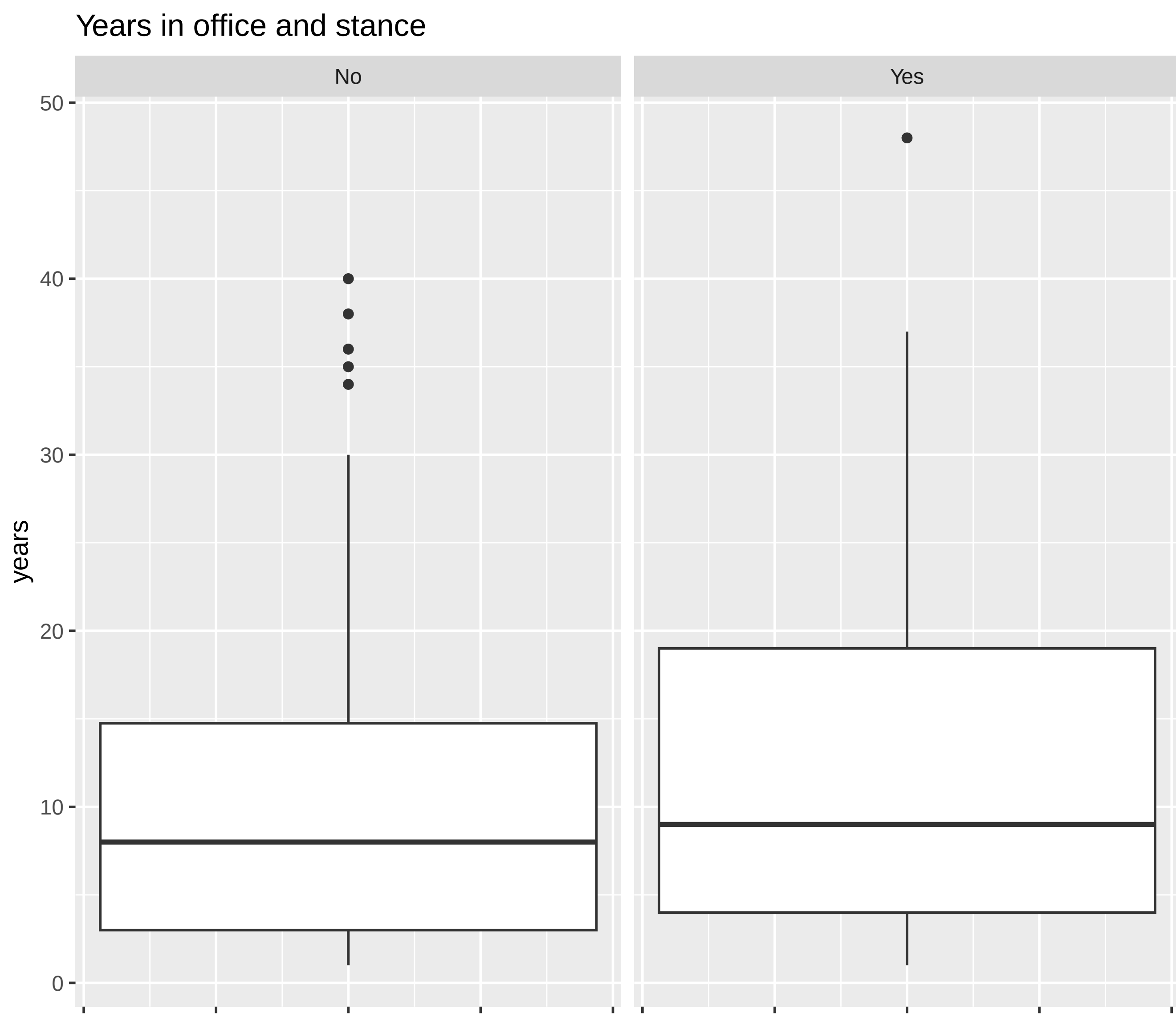

Funding supportive of PIPA/SOPA and stance
No
Yes
6e+05
4e+05
2e+05
0e+00
Money (dollars)



Stance vs Years

This plot shows the general distribution of legislators' years in office, separated by their stance on the bills.

Looking at it, it appears that those supportive of the bills have been in office for slightly longer than those opposed to them, suggesting a mild correlation. Once again, this could help explain any variation we see in our results.

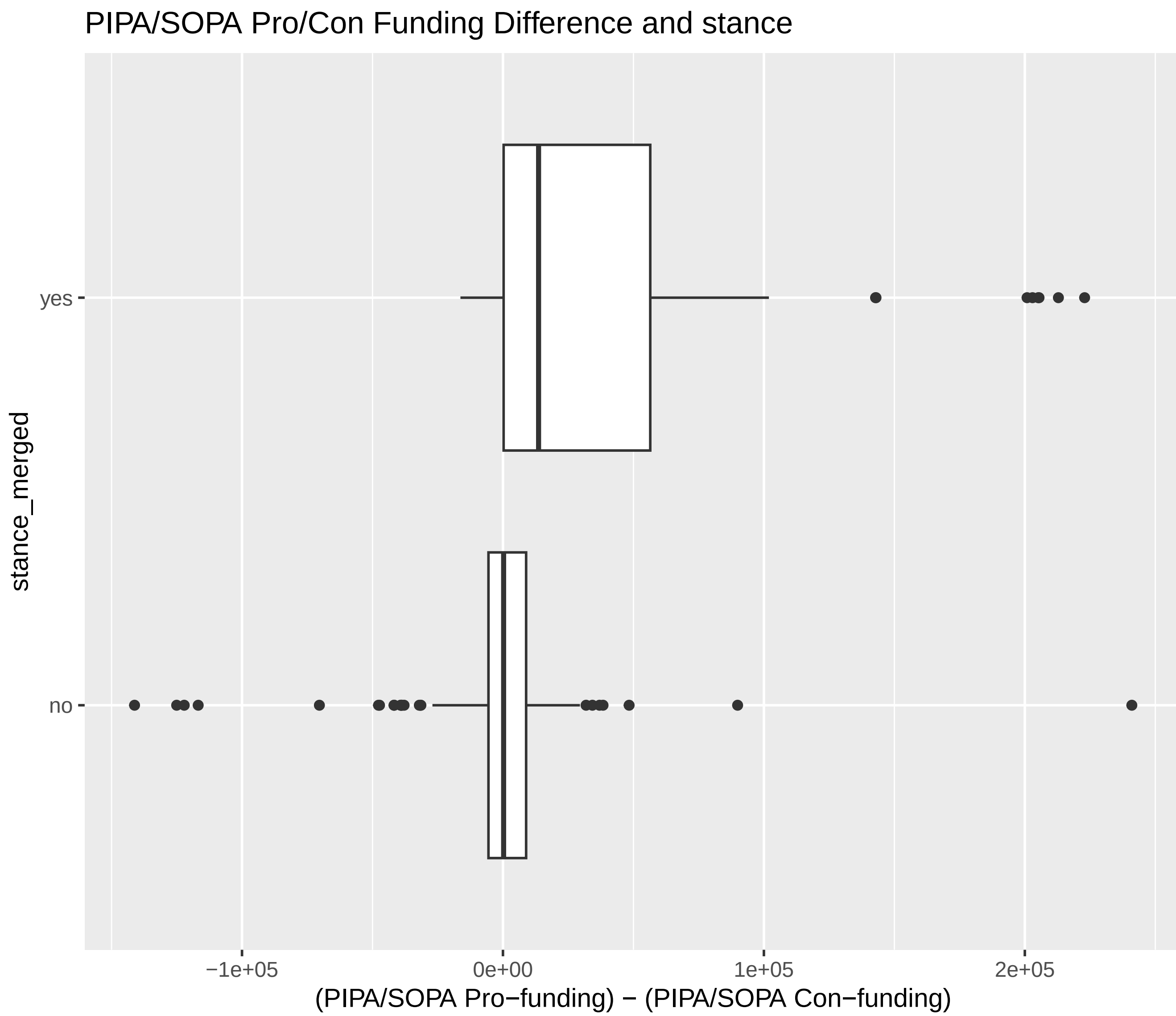


Statistical Methods

To answer our research question, we used multiple linear regression to model a relationship between stance and pro-funding vs con-funding and party affiliation.

Here we are modeling how stance relates to money-pro versus money-con. To do this, we subtract money-con from money-pro. This means that whenever the difference is positive, there is more money-pro for that stance than money-con. If the difference is negative, then there is more money-con than money-pro for that stance.

To help visualize, we have a boxplot which shows the association between the stance category and the difference between money-pro and money-con. Here you can see that there is more money-pro than money-con associated with the yes stance with the 3rd quartile extending to $54,195 ($5.4195e+04).



Next we produced a multiple linear regression model which models the relationship between stance and party affiliation and the difference of money-pro versus money-con. However, because stance is a categorical variable, we converted it to a variable called `stance_binary`. If stance was yes, stance_binary is 1. Anything else was 0. We also removed the undecided and unknown categories as we are focused on the known components of the data. We also scaled the difference to represent $10k units instead of $1 units. This gives us a much nicer numerical result.

```
## lm(formula = stance_binary ~ difference_scaled + party + difference_scaled:party,
##     data = piracy_modified)
```

The coefficients produced by our model suggests a positive slope for the funding difference, meaning more pro-funding versus con-funding is associated with a closer to yes stance. For our model this means that for every \$10k more money-pro funding than money-con funding, stance was increased by ~0.034. Remember that stance is 0 for no, or leaning no, and 1 for yes. So a positive slope means that the positive funding difference is associated with a more yes-stance. Republicans tended to lean towards opposing the legistlation while Democrats had more lean towards supporting.

```
##                              Estimate  Std. Error    t value     Pr(>|t|)
## (Intercept)               0.297927383 0.040312222  7.3904977 3.326958e-12
## difference_scaled         0.034553299 0.006176318  5.5944823 6.786331e-08
## party R                  -0.133888349 0.056087577 -2.3871302 1.785893e-02
## difference_scaled:party R -0.005816829 0.016075202 -0.3618511 7.178237e-01
```

R^2:

```
## [1] 0.1779746
```

p-value:

```
## [1] 6.786331e-08
```

Our p-value is very small here, which means that if there *truly* was no association between stance and difference, the probability of getting a slope this large or larger is extremely small. Our results suggest that if we modeled another population from another year, there would be a positive association between the funding difference and stance. This model suggests that legislators with more pro-funding relative to con-funding tended to have stance scores closer to yes. However, R^2 only explains ~18% of variation in the stance, so funding difference and party affiliation are not the only variable which is associated with stance.

Conclusion (final answer, limitations, future work, why it matters)

Overall, our analysis suggests that legislators with more pro-PIPA/SOPA funding relative to anti-PIPA/SOPA funding tended to have stance scores closer to "yes." Therefore, our main research question can be answered by saying that funding is associated with stance, but we cannot say that funding directly caused stance as this data is solely observational data.

Given that the coefficient for the `difference` variable in pro vs con funding was positive, and given that the $p-value$ is much less than α, we reject the null-hypothesis H_0, and conclude that funding is positively associated with stance.

Given that the coefficient for the `party` variable was negative, and given that the $p-value$ is much less than α, we reject the null-hypothesis H_0, and conclude that party affiliation is negatively associated with stance for republicans, and positively associated with stance for democrats.

Summary of Findings

The regression test found a positive relationship between funding difference and stance. In other words, as pro-PIPA/SOPA funding increased relative to anti-PIPA/SOPA funding, legislators tended to have stances closer to supporting PIPA/SOPA. If affiliation was Democratic then the stance is more likely to be yes, while if the affiliation was Republican then the stance was more likely to be no.

The exploratory boxplot supported this finding visually. The "yes" stance group had higher funding differences compared to the other stance groups, meaning that legislators who supported PIPA/SOPA tended to have more pro-funding relative to con-funding. The regression model confirmed this pattern statistically with a very small p-value.

However, the R^2 value was only about 18%, meaning that funding difference only explains a small amount of the variation in stance. So, while funding is related to stance, many other factors likely influence the legislators positions.

Limitations and Future Work

Here we have listed a few limitations of our methodologies:

- our model only explains ~18% of the variation in stance (funding & party are not the only predictors)
- testing our model showed that it is not viable for inference, however this is expected with a binary response variable. Our reason for not using logistic regression is that we wanted to stay aligned with what was taught in class
- Logistic regression would likely fit our data better

- Much of the data is unknown or undecided
- Missing donation values (NAs)
- Correlation $\neq$ causation

To improve upon this, we could use logistic regression models which would more accurately model the categorical response variable, and we could include data over multiple years to see more patterns in the data.